

varying from about 250,000 to above 9 million with the largest populations were included. Daily number of deaths ranged from 5 to 198. All 90 cities had daily summer O₃ measurements. Investigators conducted extensive simulation studies to test 1) the choice of the smoothing method and basic functions used to estimate the smooth function of time in the city-specific models, and 2) the number of degrees of freedom to be used in the smooth function of time. The investigators also evaluated whether each city should be assigned the same model specification or whether each city-specific model should depend on city-specific characteristics. For the former, the same degrees of freedom (ranging from 1 to 20 df/year of data) were assigned to the smooth function of time for every city. The range was determined by choosing the minimum possible degrees of freedom per year up to a maximum degrees of freedom per year that essentially removed all variation in the data beyond time scales of one week. Also, the collective experience of the investigators indicated that using more than 20 df/year does not substantially affect the risk estimates. For the latter approach, the degrees of freedom for the smooth function of time were chosen separately for each city using a fit criterion, such as the Akaike Information Criterion (AIC), or by minimizing the partial autocorrelation function (PACF) of the residuals. Nonparametric methods underestimated the standard error of the air pollution regression coefficient, penalized splines gave relatively small bias, and PACF in combination with penalized splines performed relatively well in terms of bias. Therefore, the identified risk estimate was a summer-only penalized spline estimate of respiratory mortality.

Short-term Mortality, Respiratory

In a single pollutant model, the coefficient and standard error are based on the summer-only penalized spline estimate of respiratory mortality of 0.73% (-0.39, 1.85%) per 10 µg/m³ (equivalent to 5.1 ppb) increase in O₃ from distributed lag days (Katsouyanni et al. 2009, Table 24: Distributed Lags; Penalized splines; O₃ Results).

The Health Impact Function was adjusted from the daily 1-hour max metric to the daily 8-hour max metric using a ratio of 1.13 (ratio of 1-hour max to 8-hour max ozone) (Anderson and Bell 2010, Table 2).

F.1.2 Turner et al. (2016)

Turner et al. (2016) examined the relationship between long-term O₃ exposure (1982-2004) and mortality (all-cause, cause-specific) in American Cancer Society Cancer Prevention Study-II participants (aged 30-99 years). A hierarchical Bayesian space-time model based on National Air Monitoring Stations, State and Local Air Monitoring Stations, and Community Multi-Scale Air Quality model data estimated daily eight-hour maximum ozone concentrations at the participant's address. The models considered meteorological data and levels of other ambient pollutants (PM_{2.5}, both regional and near-source, and NO₂). Turner et al. (2016) utilized Cox proportional hazard models adjusted *a priori* for individual, population subgroup, and ecological variables. Notably, the study compared annual mortality with warm-season O₃ exposures, so full-year baseline incidence rates will be used with risk estimates from this study.

Long-term Mortality, Respiratory

In a multi-pollutant model, the coefficient and standard error are based on the warm-season specific hazard ratio of 1.08 (1.06-1.11) per 10 ppb increase in seasonal average of daily 8-hour maximum O₃ concentrations (Turner et al. 2016, Table E9: Diseases of the respiratory system (cause of death), HBM O₃ (multipollutant model data, fully adjusted HR)).

F.1.3 Zanobetti and Schwartz (2008)

Zanobetti and Schwartz (2008) investigated the effects of short-term O₃ exposure on mortality (all-cause, cardiovascular, stroke, and respiratory) in an unrestricted population of children, adults, and older adults (aged 0-99 years). Between 1998 and 2000, the authors collected mortality data from the National Center for Health Statistics in 48 cities across the United States. Along with eight-hour ozone concentrations and meteorological data obtained from US EPA's Air Quality System Technology Transfer Network, the authors utilized a generalized linear model with quasi Poisson link functions to estimate the effects of short-term ozone on respiratory mortality. The model adjusted for season, day of the week, and temperature. Since ozone concentrations vary between seasons, the authors decided to restrict their analysis to ozone warm season (June - August).

Short-term Mortality, Respiratory

In a single pollutant model, the coefficient and standard error are based on the warm season excess risk estimate of 0.83% (95% CI: 0.38-1.28%) for an increase of 10 ppb in daily 8-hour mean O₃ concentrations over a summed lag structure of zero to three days (Zanobetti and Schwartz, 2008, Table 1).

The D8HourMean metric is approximated as D8HourMax in this function.

F.2 Hospital Admissions

Table F-2 summarizes the core health impact functions used to estimate the relationship between ozone and hospital admissions. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table F-2. Core Health Impact Functions for Ozone and Hospital Admissions

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
All Respiratory	Katsouyanni et al.	2009	14 U.S. Cities	65-99		D1HourMax	0.000548	0.000345	Log-linear	Warm season
All Respiratory	Katsouyanni et al.	2009	14 U.S. Cities	65-99		D8HourMax	0.000548	0.000345	Log-linear	Warm season. 8-hour max from 1-hour max using adjustment factor of 1.13, resulting in effective beta of 0.000619.

F.2.1 Katsouyanni et al. (2009)

Katsouyanni et al. (2009) used time series methods to examine the relationship between daily O₃ concentrations and hospital admissions in North America. For U.S. benefits estimation purposes, we focus on analyses performed using the U.S. hospital admission datasets. These datasets included 14 cities with populations between 291,000 and 5,377,000 between 1987-1996 with city-wide daily 1-hour maximum O₃ concentrations ranging from ~34-60 ppb. The authors used a first stage analysis protocol that used generalized linear models with either penalized or natural splines to adjust for seasonality, with varying degrees of freedom. The number of degrees of freedom were also chosen by minimizing the partial autocorrelation function of the model's residuals. Model specification approach accounted for seasonal patterns, weekend and vacation effects, and epistemics of respiratory disease. Data were also analyzed to detect potential thresholds in the concentration-response relationships. The second stage analysis used pooling approaches and assessed potential effect modification by population subgroup characteristic and indicators of the pollution mixture across study regions.

Hospital Admissions, All Respiratory (ICD-9 Codes 460-519)

In a two-pollutant model including PM₁₀, the coefficient and standard error are based on the warm season excess risk estimate of 0.28% (-0.07, 0.62%) per 10 µg/m³ (equivalent to 5.1 ppb) increase in O₃ averaged over lags 0-1 day (Katsouyanni et al., 2009, Table 40: Average of Lags 0-1 day; Penalized splines).

The Health Impact Function was adjusted from the daily 1-hour max metric to the daily 8-hour max metric using a ratio of 1.13 (ratio of 1-hour max to 8-hour max ozone) (Anderson and Bell 2010, Table 2).

F.3 Emergency Room Visits

Table F-3 summarizes the core health impacts functions used to estimate the relationship between ozone and emergency room (ER) visits. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table F-3. Core Health Impact Functions for Ozone and Emergency Room Visits

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Respiratory	Barry et al.	2018	Atlanta, GA	0-99		D8HourMax	0.00118	0.00040	Log-linear	All year
Respiratory	Barry et al.	2018	Birmingham, AL	0-99		D8HourMax	0.00118	0.00059	Log-linear	All year
Respiratory	Barry et al.	2018	Dallas, TX	0-99		D8HourMax	0.00195	0.00049	Log-linear	All year
Respiratory	Barry et al.	2018	Pittsburgh, PA	0-99		D8HourMax	0.00118	0.00040	Log-linear	All year
Respiratory	Barry et al.	2018	St. Louis, MO-IL	0-99		D8HourMax	0.00079	0.00030	Log-linear	All year

F.3.1 Barry et al. (2018)

Barry et al. (2018) investigated the effects of short-term ozone exposure on emergency department visits for respiratory disease (ICD-9 493, 786.07, 460-466, 477, 491, 492, 496, 480-486, 466.1, 466.11, 466.19) in an unrestricted population of children, adults, and older adults (aged zero-99 years) within five cities (Atlanta, GA, Birmingham, AL, Dallas, TX, Pittsburgh, PA, and St. Louis, MO-IL) across the United States. Authors obtained individual-level health data from hospitals and hospital associations in each of the five cities. Models fusing air quality monitor data with Community Multi-Scale Air Quality modeled data at 12 x 12-km grids were used to estimate ozone exposure. Barry et al. (2018) assessed associations with short-term ozone exposure with daily eight-hour maximum ozone concentrations. The authors implemented Poisson log-linear models to estimate risk values with three day moving averages.

ER Visits, Respiratory (ICD-9 Codes 493, 786.07, 460-466, 477, 491, 492, 496, 480-486, 466.1, 466.11, 466.19)

In single-pollutant models, the coefficient and standard error are based on rate ratios of 1.03 (95% CI: 1.01-1.05) in Atlanta, GA, 1.03 (95% CI: 1.00-1.06) in Birmingham, AL, 1.05 (95% CI: 1.02-1.07) in Dallas TX, 1.03 (95% CI: 1.01-1.05) in Pittsburgh, PA, and 1.02 (95% CI: 1.01-1.04) in St. Louis, MO-IL for an increase of 25 ppb in full-year 8-hour daily maximum O₃ concentrations (three day moving average) (Barry et al. 2018, Table 3).

F.4 Minor Effects

Table F-4 summarizes the core health impacts functions used to estimate the relationship between ozone and minor effects. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table F-4. Core Health Impact Functions for Ozone and Minor Effects

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
School Loss Days, All Cause	Gilliland et al.	2001	Southern California	5-18		D8HourMax	0.007824	0.004445	Log-linear	All year, 8-hour max from 8-hour mean.
Minor Restricted Activity Days	Ostro and Rothschild	1989	Nationwide	18-64	PM _{2.5}	D1HourMax	0.002200	0.000658	Log-linear	
Minor Restricted Activity Days	Ostro and Rothschild	1989	Nationwide	18-64	PM _{2.5}	D8HourMax	0.002200	0.000658	Log-linear	8-hour max from 1-hour max using adjustment factor of 1.14, resulting in effective beta of 0.002508.
Hay Fever/Rhinitis	Parker et al.	2009	Nationwide	3-17	PM _{2.5} , PM _{2.5-10} , NO ₂ , SO ₂	Annual (D24HourMean)	0.01655	0.00390	Logistic	Warm season; long term